令和７年度
前期日程

英　　語

教育学部・地域科学部・医学部医学科・
医学部看護学科・工学部・社会システム経営学環

問題冊子

注意事項

(1) 試験開始の合図があるまで，問題冊子を開かないこと。

(2) 問題冊子は22ページで，解答用紙は3枚である。問題冊子や解答用紙に，落丁，乱丁，印刷不鮮明のものがあった場合は，ただちに試験監督者に申し出ること。

(3) 受験番号は，3枚の解答用紙のそれぞれの指定箇所に**丁寧に，読みやすい字で**記入すること。

(4) **問題は，Section 1 から Section 5 までの5つからなる。**

(5) 解答は，解答用紙の指定箇所に**丁寧に，読みやすい字で**記入すること。

(6) 解答用紙は，持ち帰らないこと。

(7) 問題冊子は，持ち帰ること。

(8) 複数の Section の満点に対する配点の比率(%)を次のとおりとする。
Section 1 から Section 3 の合計は 50 %，Section 4 と Section 5 の合計は 50 %

(9) 設問は**英文**で書かれている。

Section 1

Questions 1–10 :

Read the following text and fill in the blanks (1)–(10) to complete the sentences. For each blank, you have four choices given below. Choose the correct option and write A, B, C or D in boxes 1–10 on your answer sheet.

In our everyday lives, time is a precious commodity. We can gain ___(1)___ lose it. We can save, spend or waste it. If our crimes are revealed, we risk having to do time.

To scientists, time is something we ___(2)___ measure. Clocks have, over the centuries, been the high tech artifacts of their era — the water clock, the pendulum clock, Harrison's chronometer, and so ___(3)___ up to the incredible precision of atomic clocks — marvels of modern technology, albeit without the evident aesthetic quality of more traditional timepieces. (Though engineering friends tell me that, ___(4)___ through a microscope, there's beauty in the intricacies of a silicon chip.)

Before there was a reliable calendar — or any records, or artifacts that could be reliably dated — the past was a 'fog'. But this didn't stop efforts to ___(5)___ fanciful precise chronologies. Most precise of all was that worked out by James Ussher, Archbishop of Armagh, according to ___(6)___ the world began at 6 pm on Saturday, 22 October, 4004 B.C. Right up until 1910, bibles published by Oxford University Press displayed Ussher's chronology alongside the text.

Even in the 17th-century, Ussher's estimates ran into problems. Jesuit missionaries returned ___(7)___ China, telling of detailed historical records dating back to dynasties before 2350 B.C. — the proclaimed date for Noah's Flood. Many were sceptical that the entire history of Earth's mountains, rivers and fauna could be squeezed into 6000 years. Sir Isaac Newton, in his old age, ___(8)___ abandoned science, and was obsessed with completing his own 'Chronology of Ancient Kingdoms'. He ___(9)___ not contest Ussher's dating of human origins, but conjectured that the six 'days' of Genesis could each be a prolonged era.

In the 19th-century, Darwin's genius was to recognise how "natural selection of favoured variations" could have transformed primordial life into the amazing varieties of creatures, now mainly extinct, that have crawled, swum or flown on Earth. But this emergence — a higgledy-

piggledy process, proceeding ___(10)___ any guiding hand — is inherently very slow. Darwin guessed that evolution required not just millions but hundreds of millions of years. He was mindful of supporting evidence from geology. He estimated, by an argument that was actually flawed (and which he cut from later editions of his book), that to carve out the Weald of Kent took 300 million years. If he had seen the Grand Canyon he could have made a more convincing estimate.

(Adapted from Martin Rees "We Still Don't Fully Understand Time," *TIME*, https://time.com/6969823/martin-rees-time/)

<Options>

(1)	**A.** and	**B.** or	**C.** to	**D.** too
(2)	**A.** are	**B.** aren't	**C.** can	**D.** can't
(3)	**A.** after	**B.** at	**C.** forth	**D.** with
(4)	**A.** view	**B.** viewed	**C.** viewer	**D.** viewing
(5)	**A.** impose	**B.** imposed	**C.** imposer	**D.** imposing
(6)	**A.** after	**B.** since	**C.** until	**D.** which
(7)	**A.** by	**B.** for	**C.** from	**D.** in
(8)	**A.** had	**B.** is	**C.** may	**D.** would
(9)	**A.** did	**B.** has	**C.** is	**D.** was
(10)	**A.** at	**B.** in	**C.** of	**D.** without

Section 2

Questions 11–20 :

Read the following text, choose from A–K the option that best fits each blank (11)–(20) and write the letter on your answer sheet. There is one extra option that is not used.

A Societal Mission

Stanford was founded almost 150 years ago on a bedrock of societal purpose. Our mission is to contribute to the world by educating students for lives of leadership and contribution with integrity; advancing fundamental knowledge and cultivating creativity; leading in pioneering research for effective clinical therapies; and ___(11)___.

Who we are

What drives our faculty, students, and staff is a spirit of possibility, a legacy inherited from our pioneering founders who created Stanford in the memory of their only child to benefit other people's children. At Stanford, we learn from our past but ___(12)___ to make the world a better place.

We believe strongly in the role that higher education plays in supporting a thriving society. Colleges and universities not only prepare students for their chosen path, but they help create the next generation of citizen leaders and problem solvers. Higher education institutions pursue a distinct and enduring mission in our society, dedicated solely to the creation and dissemination of knowledge. The breadth and richness of the fabric that comprises the world's institutions of higher learning are ___(13)___.

Educating tomorrow's global citizens

Stanford provides students the opportunity to engage with big ideas, to cross conceptual and disciplinary boundaries, and to become global citizens who embrace diversity of thought and experience. We offer students broad and deep academic programs across seven schools and multiple fields — including the arts and humanities, natural and social sciences, engineering, sustainability, medicine, law, education, and business. The university's breadth of excellence and culture of innovation uniquely position it to ___(14)___, who offer students the knowledge and tools to discover and embrace new ideas, and to prepare for successful careers and lives of service. The need for strong ethical principles to achieve a common good is embedded across our educational offerings. Stanford's distinguished alumni network

and highly engaged alumni community provide students with ___(15)___, who help to create partnerships with the world beyond campus, once again enhancing the collective contributions of our community.

Creating knowledge in an environment of academic freedom

Unfettered inquiry is the foundation of research and education. Driven by a deep belief in academic freedom, we ___(16)___ and develop new ideas to advance knowledge. Fundamental, curiosity-driven research is central to the university's mission, and students have extensive opportunities to join Stanford scholars in research that develops new knowledge and deepens understanding of ourselves and the world around us. A hallmark of Stanford is our extensive and vibrant ecosystem of interdisciplinary research. With all seven of Stanford's schools located on our historic campus and many institutes serving as a hub for collaboration across academic fields, ___(17)___ are numerous and the results are evident. Stanford has had 36 Nobel Prize winners since the university's founding and is tied among research universities for the most Nobel Prizes won since 2000. We are equally purposeful in thinking actively about ___(18)___ in the broader world.

Rapidly advancing knowledge into application

Building on our culture of interdisciplinary collaboration that drives discovery and innovation, the university is creating "accelerators" to readily support university researchers with funding, infrastructure, technological resources, and external partners so that discoveries made at Stanford can ___(19)___.

Examples include the Innovative Medicines Accelerator, dedicated to reducing the time and cost of making medicines while ensuring more effective treatments are reaching patients, and the Sustainability Accelerator within the Stanford Doerr of Sustainability, the university's first new school in 70 years and a testament to its commitment to being a leader on climate-related research and education. Work is also ongoing within Stanford's Graduate School of Education to accelerate more effective and equitable learning solutions, and at the Stanford Impact Labs, ___(20)___ and development pipeline for the social sciences.

(Adapted from https://www.stanford.edu/about/)

<Options>

A. accelerating solutions and amplifying their impact

B. attract and retain the best faculty in the world

C. become solutions with impact far beyond campus

D. continually pose questions

E. focus on the future, continually looking for new ways

F. integral to long-term human advancement and well-being

G. lifelong access to mentors and peers

H. selected a rigorous academic program

I. the ethical implications of our research and its application

J. the opportunities for disruptive breakthroughs

K. which creates an innovative research

The examination continues on the next page.

Section 3

Questions 21–30 :

Read the following text and answer the questions that follow. Write the correct letter in boxes 21–30 on your answer sheet.

11 Foolproof Ways to Start a Conversation

Striking up a conversation — especially with a stranger — is a lot like adding kindling to a fire pit and hoping it ignites. Choose the wrong starter, and the flame will ___(21)___. But when you get it right, conversation and connection can spark.

Conversations play an essential role in our well-being. Having just one quality talk a day, especially face-to-face, can increase happiness and lower stress levels by the end of the day, research has found. You don't even have to know the person. In one study, people overestimated how awkward deep, meaningful talks with strangers would be, and found that instead, these conversations uplifted them and made them feel more connected than they expected. Plus, asking questions makes people like us more, which can be a boon to self-esteem.

But what should you say to get more than a one-word answer in response? We asked people who excel at small talk to share their favorite conversation-starters.

"What's the best dining experience you've ever had?"

Raele Altano, a communication coach in New York City, likes the "___(22)___" approach to conversation-starters. It stands for hobbies, entertainment, food, and environment (meaning the setting you're in). Those four universal pillars lend themselves to talking points in almost every situation, she says. Who doesn't love chatting about food, for example? Asking about favorite restaurants "helps you learn something new and get to know them and their preferences," she says. Altano recently asked someone about their best recent dining experience and what made it so great — and learned that the other person had traveled to Japan on ___(23)___ she did for hers, and that they were at the same restaurant in Tokyo only days apart. That forged a bond they otherwise never would have known existed, she says.

"What's been the best part of your week so far?"

Asking someone how they are is practically a reflex. Instead, switch things up by asking the next person you encounter about the highlight of their week, suggests Robin Shear, the Detroit-based author of *Messy Joy: How Joy Can Begin Before Your Difficulties End.* "People are caught off guard because they're surprised someone cares enough about them to ask," she says. "They're usually jolted out of the mundane and suddenly feel important." When she asks people this question, she typically finds they have to pause to consider it — and they then tell her that they'll have to think about the bright spots of their week more often. A grocery store checkout clerk, for example, recently told Shear about ___(24)___. By the end of the conversation, "We both had tears rolling down our cheeks," she recalls. "I'll never forget it."

"What's the nicest thing anyone's ever said about you?"

Every time Shear asks someone this question, they smile. "___(25)___," she says. Take this recent exchange at a fast-food joint: The guy waiting on her lit up and said, "Someone once told me I'm a people person." That led to a conversation about how he might use his people skills in the future, and his hope to eventually become a teacher. "I thanked him for being the best part of my day," Shear recalls. And, by the way: The nicest thing anyone ever said about her came from her daughter, who told her, "You know how to make people feel like they matter."

"[(26)]"

You read that right — and your reaction is exactly why Tenyse Williams likes this question. "It always grabs people off-guard," she says. Plus, it requires imagination, curiosity, and introspection; not to mention, it's fun. Williams, who's the founder of a marketing agency in Brooklyn, once asked a chef this question during a large speaking event. After thinking it over, the woman responded: "Definitely a time traveler, to snatch all the historic recipes before anyone else." The room dissolved into laughter, and the lighthearted moment buoyed the rest of the evening.

"What's the most interesting thing you've learned recently?"

This question is open-ended and inviting — and can lead to fascinating follow-ups, says Erica Thomas, who hosts supper club events in Atlanta and is the founder of the website Eating With Erica. "It works wonders at networking events, parties, and even when you're simply chatting with friends or acquaintances," she says. Thomas especially likes that it draws people into discussions about their passions and curiosities, helping foster an ___(27)___ ideas. When she used it at one of her dinner parties, for example, it prompted a discussion about travel to Europe — which led to deeper conversations about life lessons, change, and spontaneity.

"That belt pairs so nicely with your outfit! [(28)]"

Everyone loves a compliment: Research suggests receiving praise is as thrilling as a monetary reward. That's why Jillian Amodio, a licensed social worker in Annapolis, Md., starts conversations by dispensing kind words, followed by a relevant question. Tried-and-true favorites include: "I love that shade of blush. Where did you get it?" and "You're great at your job. Is this a field you've always wanted to be in?"

Amodio, who teaches career development at Anne Arundel Community College, will sometimes use this line with her students: "You're always the first to offer an answer in class. I admire that about you! Have you always been this outgoing?" Offering a compliment helps people "feel seen and valued," she says — and following-up with a question ensures a more expanded conversation than a simple thank you or smile.

"If you could do anything you want without having to worry about money, what would you want to do and why?"

Ah, the million-dollar question. We all squirrel away dreams of what we'd do if we suddenly came into a lot of money, Woo says, and this wording gives people permission to set aside real-life concerns and keep things fun and upbeat. "It provides a wide canvas for the person to illustrate their dreams and aspirations," she says. Woo recalls using this question as an icebreaker, and noticing the room became louder and more lively with laughter. People discovered they shared common interests — there were even aspiring magicians in the room — and found each other afterward to chat more.

"What's your perfect Saturday?"

People seem to enjoy answering this question, says Meg Irvin, who works at a communications firm in Richmond, Va. If they turn it around and ask her about her perfect weekend, she tells them she has two young kids, so "___(29)___." Also on the docket: popping into a farmer's market, taking a walk in the sunshine, and maybe checking a day trip off the running list of destinations she keeps on her phone. Who knows? Maybe you and your conversation partner will make plans to share a Saturday activity together.

"How would your best friend introduce you?"

Ask Jenn Whitmer this question, and she'll tell you her BFFs would describe her as bold, energetic, and expressive. "Full of joy and ready to give that to others," she says. You might also learn that she'll ___(30)___ song at the drop of a hat and has a contagious laugh. Whitmer, a St. Louis-based keynote and TEDx speaker who hosts the Joyosity podcast, likes this question's revealing nature, and the way it demonstrates a genuine desire to learn about the person you're talking to. "Stories come out, and the follow-up questions are so easy because you're connecting over something real," she says. "Most people dread small talk because it feels meaningless or transactional. People want someone who's actually interested in them as a human." Asking great questions — and listening to the answer — can help anyone master the art of conversation, she says.

(Adapted from Angela Haupt "11 Foolproof Ways to Start a Conversation," *TIME*, https://time.com/6976468/conversation-starters-small-talk/)

Q 21 : Choose from A–E the option that best fits the blank in (21), and <u>write A, B, C, D or E</u> in box 21 on your answer sheet.

A. account for

B. break into

C. exchange of

D. fizzle out

E. see through

Q 22 : Choose from A–E the option that best fits the blank in (22), and write A, B, C, D or E in box 22 on your answer sheet.

A. FHEF

B. HEFE

C. NATURE

D. OECD

E. QUICK

Q 23 : Choose from A–E the option that best fits the blank in (23), and write A, B, C, D or E in box 23 on your answer sheet.

A. every single person

B. it's never true

C. sleeping in sounds pretty glorious

D. the weekly dinner she had just had with her son

E. their honeymoon at the same time

Q 24 : Choose from A–E the option that best fits the blank in (24), and write A, B, C, D or E in box 24 on your answer sheet.

A. every single person

B. it's never true

C. sleeping in sounds pretty glorious

D. the weekly dinner she had just had with her son

E. their honeymoon at the same time

Q 25 : Choose from A–E the option that best fits the blank in (25), and write A, B, C, D or E in box 25 on your answer sheet.

A. Every single person

B. It's never true

C. Sleeping in sounds pretty glorious

D. The weekly dinner she had just had with her son

E. Their honeymoon at the same time

Q 26 : Choose from A–E the most appropriate title for (26), and write A, B, C, D or E in box 26 on your answer sheet.

A. How did you get to know your partner?

B. How did you learn to be so creative with your accessories?

C. What's your ideal successor like?

D. Would you rather be a time traveler or a mind reader?

E. Would you tell us about your dreams when you were a child?

Q 27 : Choose from A–E the option that best fits the blank in (27), and write A, B, C, D or E in box 27 on your answer sheet.

A. account for

B. break into

C. exchange of

D. fizzle out

E. see through

Q 28 : Choose from A–E the most appropriate title for (28), and write A, B, C, D or E in box 28 on your answer sheet.

A. How did you get to know your partner?

B. How did you learn to be so creative with your accessories?

C. What's your ideal successor like?

D. Would you rather be a time traveler or a mind reader?

E. Would you tell us about your dreams when you were a child?

Q 29 : Choose from A–E the option that best fits the blank in (29), and write A, B, C, D or E in box 29 on your answer sheet.

A. every single person

B. it's never true

C. sleeping in sounds pretty glorious

D. the weekly dinner she had just had with her son

E. their honeymoon at the same time

Q 30 : Choose from A–E the option that best fits the blank in (30), and write A, B, C, D or E in box 30 on your answer sheet.

A. account for

B. break into

C. exchange of

D. fizzle out

E. see through

The examination continues on the next page.

Section 4

Questions 31–40 :

Read the following text and answer the questions that follow. Write the correct letter in boxes 31–40 on your answer sheet.

Scientists Say They've Discovered A 'Phonetic Alphabet' in Whale Calls

Scientists have accomplished a whale of a feat. They've identified previously unknown complexity in whale communication by analyzing thousands of recorded sequences of sperm whale clicks with artificial intelligence.

Variations in tempo, rhythm and length of the whales' click sequences, called codas, weave a rich acoustic tapestry. These variables hint that whales can combine click patterns in multiple ways, mixing and matching phrases to convey a broad range of information to one another.

What sperm whales are saying with their clicks remains a ___(31)___ to human ears. Still, uncovering the scope of whales' vocal exchanges is an important step toward linking whale calls to specific messages or social behaviors, the scientists reported May 7 in the journal Nature Communications.

"This work builds on a lot of ___(32)___ focused on understanding the calls of sperm whales. However, this is the first work that has started to look at sperm whale calls in their wider communicative context and in the context of exchanges between whales, which has made some of the findings possible," said study coauthor Dr. Daniela Rus, director of the Computer Science and Artificial Intelligence Laboratory (CSAIL) at MIT, in an email.

"Understanding what aspects of their codas they can control and ___(33)___ how they can encode information in their calls," Rus said.

The researchers dubbed their catalog of sound combinations a "phonetic alphabet" for sperm whales, comparing variations in the whales' click sequences to the production of different phonetic sounds in human speech.

But while the team's findings are interesting, that term offers a misleading perspective on whales' vocal interactions, said Dr. Luke Rendell, a researcher at the University of St. Andrews in the United Kingdom whose work focuses on communication in marine mammals, in an email.

"The presentation of the 'phonetic alphabet' — it's nothing of the sort," said Rendell, who was not involved in the research.

"The way the tempo variation is used is completely different to how, ___(34)___ construct linguistic expression," he said. "There's no evidence of that, and it's not a super helpful interpretation because it forces everything into a restricted and somewhat over-sold perspective of 'is it like human language or not,' when there are a much broader range of interpretations available."

Pattern recognition

Sperm whales produce their clicks by forcing air through an organ in their heads called the spermaceti, and these sounds can be as loud as 230 decibels — louder than a rocket launch and capable of rupturing human eardrums — another team of scientists previously reported in the journal *Scientific Reports*.

For the new study, the researchers ___(35)___ patterns in audio data collected by The Dominica Sperm Whale Project, a repository for observations of sperm whales that inhabit the Caribbean Sea. The recordings represented the voices of approximately 60 sperm whales — a subset of a group of about 400 whales known as the Eastern Caribbean clan — and the vocalizations were recorded between 2005 and 2018.

Prior research had identified 150 types of codas in sperm whales worldwide, but the Caribbean whales used just 21 of those codas.

The scientists examined the timing and frequency of 8,719 coda sequences — in solitary whale utterances, in choruses and in call-and-response exchanges between whales. When visualized with artificial intelligence, ___(36)___.

The study authors defined four features in codas: rhythm, tempo, rubato and ornamentation. Rhythm describes the sequence of intervals between clicks. Tempo is the duration of the entire coda. Rubato refers to variations in duration across adjacent codas of the same rhythm and tempo. And ornamentation is an "extra click" added at the end of a coda in a group of shorter codas, Rus explained.

These so-called ornament clicks "occur more towards the beginning and end of turns" during vocal exchanges between whales, "behaving like discourse markers," Rus said.

The discovery that whales could synchronize variations in coda tempo was "a really interesting observation," Rendell said.

"I am less convinced by the 'ornamentation'," he added. "It occurs very rarely, and I think we need more evidence that they aren't just production glitches," or filler sounds, "___(37)___"

In all, the program detected 18 types of rhythm, five types of tempo, three types of

rubato and two types of ornamentation. These coda features could all be mixed and matched to form an "enormous repertoire" of phrases, the study authors reported. What's more, meaning could be tweaked even further depending on the placement of a coda — following or overlapping other codas — within an exchange or chorus involving two or more whales.

Interactive experimentation

"Actually, many of us have been waiting for advanced technology to allow us to do something like this for decades!" said Dr. Brenda McCowan, a professor at the University of California Davis School of Veterinary Medicine, in an email.

McCowan, who was not involved in the research, was part of another team that, in 2021, conducted an interactive "___(38)___" with a humpback whale in waters near Alaska. For about 20 minutes, a curious whale repeatedly responded to a recording of a humpback song transmitted from the scientists' boat.

"This particular playback (with the humpback in 2021) was an opportunistic experiment with an inquisitive whale engaging us both behaviorally and vocally, and completely at her own volition," McCowan said.

Such interactive experimentation with whales, along with observations of whale behavior, could be an important part of unraveling the syntax of sperm whale click sequences, the authors wrote in the study.

Their machine learning method may also prove useful for ___(39)___, McCowan added.

"Tempo, rhythm, rubato and ornamentation are likely to be found in other species of whales," McCowan said. "We already know this is true of humpback song. But there is also evidence for this type of patterning in other aquatic, terrestrial and arboreal species to which this approach could be applied."

But although this technique is helpful for ___(40)___, it's no Rosetta Stone, Rendell cautioned.

"Machine learning is great for finding patterns in large datasets," he said, "but it doesn't create meaning."

(Adopted from Mindy Weisberger "Scientists say they've discovered a 'phonetic alphabet' in whale calls," *CNN*, https://edition.cnn.com/2024/05/13/world/whale-communication-coda-alphabet-scn/index.html)

Q 31 : Choose from A–E the option that best fits the blank in (31), and write A, B, C, D or E in box 31 on your answer sheet.

A. behavior

B. genome

C. mystery

D. precision

E. science

Q 32 : Choose from A–E the option that best fits the blank in (32), and write A, B, C, D or E in box 32 on your answer sheet.

A. harsh conditions

B. images of its landing site

C. periods of darkness

D. prior work

E. surprising teams

Q 33 : Choose from A–E the option that best fits the blank in (33), and write A, B, C, D or E in box 33 on your answer sheet.

A. find social behaviors of

B. heat up to damage

C. make the most of

D. sleep because it was not

E. vary helps us understand

Q 34 : Choose from A–E the option that best fits the blank in (34), and write A, B, C, D or E in box 34 on your answer sheet.

A. as well as where, their relationship over

B. for example, it finds alternatives with

C. say, we use elements of an alphabet to

D. though, that is not a coincidence with

E. too, whales around the world started to

Q 35 : Choose from A–E the option that best fits the blank in (35), and write A, B, C, D or E in box 35 on your answer sheet.

A. began the removal of

B. recruited assistants for

C. served the same sentence for

D. studied the way we compose sentence

E. used machine learning to detect

Q 36 : Choose from A–E the option that best fits the blank in (36), and write A, B, C, D or E in box 36 on your answer sheet.

A. previously unseen coda patterns emerged

B. rules and patterns never became clear

C. the loud noise came to be used

D. the origins of the word grammar disappeared

E. whales understood how their language works

Q 37 : Choose from A–E the option that best fits the blank in (37), and write A, B, C, D or E in box 37 on your answer sheet.

A. between grammar and glamour illustrates.

B. for the study of the mysterious noise shows.

C. just when they come to study.

D. like when we say 'um' or 'err.'

E. where surroundings of someone or something.

Q 38 : Choose from A–E the option that best fits the blank in (38), and write A, B, C, D or E in box 38 on your answer sheet.

A. awareness

B. conversation

C. eyesight

D. footsteps

E. knowledge

Q 39 : Choose from A–E the option that best fits the blank in (39), and write A, B, C, D or E in box 39 on your answer sheet.

A. how widespread such complex behavior may have been

B. organizing a group of humpback whales

C. studying other types of animal vocalizations

D. the evidence that other whales that went extinct a long time ago

E. understanding the meaning of the white sounds

Q 40 : Choose from A–E the option that best fits the blank in (40), and write A, B, C, D or E in box 40 on your answer sheet.

A. approaching day-to-day existence on planet earth

B. creating a means of communication with whales

C. growing out of a need to communicate

D. identifying certain aspects of communication

E. making communication easier between different species

Section 5

Questions 41–50 :

Read the following text, then rearrange the underlined words in (41)–(50) to complete each expression. In each question, there is ONE word that is not used. Then, in boxes 41–50 on your answer sheet, write the word that is not needed to complete the sentence.

Example:

I think that (0) a, has, have, pen, she

The correct expression is **She has a pen**.

→ **have** is not used. → So, write **have** in Box 0.

0
have

Write ONLY the unnecessary word in the boxes on your answer sheet.

Cicadas spend the vast majority of their lifetime — more than 90% — underground. But this spring, two broods of more than a trillion cicadas (41) above, debut, make, not, soil, the, their, will across the Midwest and Southeast in an event that (42) centuries, happened, has, in, makes, more, not, than, two.

"When these emerge, it really is a unique, natural phenomenon," says PJ Liesch, the director of the University of Wisconsin-Madison Insect Diagnostic Lab, (43) compares, emergence, insects', the, to, under, who that of Monday's total solar eclipse. "If you think about your entire lifetime, you might only have a few opportunities ever (44) able, an, be, eclipse, see, to, to, with. And for these periodical cicadas, it's very similar."

Cicadas have already begun to emerge in some Southern states including North Carolina and Georgia. The insects, (45) been, have, into, living, underground, which for more than a decade, begin to dig upwards (46) around, depart, reaches, soil, temperature, the, when 64

degrees Fahrenheit or more.

"Once that happens, you often have a decent rain shower, that can (47) a, be, cause, emergence, rain, this, trigger, to. So that's what we're starting to see in some parts of the U.S. For us in Northern Illinois and in southern Wisconsin it's probably going to be another month, month and a half (48) activity, before, bulk, of, our, see, the, visible, we at more northern parts of the cicada range."

Here's what to know about the cicadas.

What are the two broods of cicadas emerging?

There are two adjacent broods of cicadas (49) as, at, be, emerging, same, that, the, time, will. Brood XIX reside in Alabama, Arkansas, Georgia, Kentucky, Louisiana, Missouri, Mississippi, North Carolina, Oklahoma, South Carolina, Tennessee and Virginia. Brood XIII could be seen in Iowa, Wisconsin and possibly even Michigan.

What do cicadas do?

Cicadas have a lengthy lifespan of either 13 or 17 years, depending on the brood. There are three species of 17-year cicadas, and four species of 13-year cicadas. The insects (50) a, almost, at, depth, done, entire, live, lives, soil, their of anywhere from eight inches or more, according to Liesch. During the time they are burrowed underground, they typically tunnel around and feast on the sap of tree roots as a source of food.

(Adapted from Solcyré Burga "More Than a Trillion Cicadas Are Coming. Are You Ready?" *TIME*, https://time.com/6966406/cicadas-trillion/)